

TX4 Robustness Statistical Appendix

Exact and tiered statistical outputs

A. Master-table dimensions

The master table contains 329,440 portfolio rows over 20,590 conditions. Exact H4 rows: 20,590; exact all-portfolio calibration rows: 6,816; surrogate proper-subset rows: 302,460.

B. QMC summary

campaign endpoint n_conditions success proportion ci95_low ci95_high interpretation QMC H4_PRESENT 4096
2367.0 0.577881 0.562689 0.592926 bounded-box engineering probability; proper subsets outside calibration are
surrogate-tier QMC EXACT_H4 4096 301.0 0.073486 0.065889 0.081883 bounded-box engineering probability;
proper subsets outside calibration are surrogate-tier QMC NONCOMPOSABLE 4096 0.0 0.000000 0.000000
0.000937 bounded-box engineering probability; proper subsets outside calibration are surrogate-tier QMC
delta_H4_mean 4096 NaN 0.028325 -0.095992 0.225307 empirical 2.5/97.5 percentiles

C. MC summary

campaign endpoint n_conditions success proportion ci95_low ci95_high interpretation MC H4_PRESENT 5000
2907.0 0.581400 0.567668 0.595007 bounded-box engineering probability; proper subsets outside calibration are
surrogate-tier MC EXACT_H4 5000 356.0 0.071200 0.064396 0.078662 bounded-box engineering probability;
proper subsets outside calibration are surrogate-tier MC NONCOMPOSABLE 5000 0.0 0.000000 0.000000
0.000768 bounded-box engineering probability; proper subsets outside calibration are surrogate-tier MC
delta_H4_mean 5000 NaN 0.029846 -0.092035 0.218205 empirical 2.5/97.5 percentiles

D. Method comparison

campaign	n	both_stable	full_only	mode_only	neither	mcnemar_p	status	MC	5000	823	0	1270	2907	0.000000e+00
PAIRED_MODE_COMPARISON	QMC	4096	743	0	986	2367	3.058118e-297	PAIRED_MODE_COMPARISON						

E. Sobol summary

endpoint parameter S1 ST N_base status deviation H4_PRESENT g 0.004659 0.039040 1024
SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations H4_PRESENT
k 0.510143 0.699922 1024 SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE
evaluations H4_PRESENT t 0.024651 0.054318 1024 SURROGATE_SALTELLI indices use calibrated surrogate,
not fresh exact Saltelli DAE evaluations H4_PRESENT h 0.003227 0.022243 1024 SURROGATE_SALTELLI
indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations H4_PRESENT epsilon 0.492282
0.634389 1024 SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations
H4_PRESENT damping 0.001464 0.001925 1024 SURROGATE_SALTELLI indices use calibrated surrogate, not
fresh exact Saltelli DAE evaluations H4_PRESENT inertia 0.010765 0.025557 1024 SURROGATE_SALTELLI
indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations EXACT_H4 g 0.083624 0.263662 1024
SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations EXACT_H4 k
-0.040083 0.869212 1024 SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE
evaluations EXACT_H4 t -0.002415 0.183221 1024 SURROGATE_SALTELLI indices use calibrated surrogate,
not fresh exact Saltelli DAE evaluations EXACT_H4 h 0.028033 0.040984 1024 SURROGATE_SALTELLI indices
use calibrated surrogate, not fresh exact Saltelli DAE evaluations EXACT_H4 epsilon -0.221062 0.747133 1024
SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact Saltelli DAE evaluations EXACT_H4
damping 0.000464 0.021851 1024 SURROGATE_SALTELLI indices use calibrated surrogate, not fresh exact
Saltelli DAE evaluations EXACT_H4 inertia 0.004287 0.071698 1024 SURROGATE_SALTELLI indices use
calibrated surrogate, not fresh exact Saltelli DAE evaluations

F. Logistic summary

endpoint status parameter coefficient odds_ratio n deviation H4_PRESENT FIT g -1.270247 0.280762 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT k 5.262715 193.004856 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT t -4.610627 0.009946 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT h -0.334342 0.715809 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT epsilon -5.730459 0.003246 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT damping -0.235769 0.789964 5000 predictive logistic fit; no causal interpretation H4_PRESENT FIT inertia 1.901780 6.697806 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT g -0.376294 0.686401 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT k -0.176499 0.838199 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT t 0.780460 2.182476 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT h -0.197558 0.820733 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT epsilon -0.455609 0.634062 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT damping -0.019448 0.980740 5000 predictive logistic fit; no causal interpretation EXACT_H4 FIT inertia 0.617434 1.854165 5000 predictive logistic fit; no causal interpretation

G. File inventory

Master CSV, compressed CSV, Parquet, 1D sweeps, 2D phase Parquet, QMC/MC summaries, Morris, Sobol, logistic, g-star, method, return, mode, TDS, Julia, claim matrix, metadata, and headline JSON are retained under results/.

H. Audit limitations

No physical uncertainty weights, physical robust radius, fresh random Julia parity, fresh same-model PowerDynamics network parity, nonlinear TDS, EMT, current-limit, DC-link, protection, or hardware certification is claimed.